

Tropical Meteorology



Chapter 5: Mesoscale Systems in the Tropics



Objective

- *To Define Mountain wave*
- *To Know The lee waves and Related clouds*
- *To Identify the land and sea breezes*
- *To Distinguish anabatic and katabatic winds*

Mountain wave and lee waves

- Mesoscale systems are atmospheric phenomena with spatial scales of $\sim 2\text{--}200$ km and time scales of hours to a day.

Mountain Waves

- Mountain waves refer to the entire phenomenon of atmospheric waves generated by the flow of air over a mountain barrier.
- When stable air is forced to rise over a mountain range, it creates oscillations in the atmosphere, forming a series of waves with crests and troughs.
- Mountain waves can extend both vertically and horizontally downstream of the mountains and can have significant impacts on weather, aviation, and atmospheric dynamics.
- They can produce clear air turbulence, induce the formation of clouds like lenticular clouds, and affect local weather patterns downwind of the mountains.
- Mountain waves can also influence the dispersion of pollutants and play a role in the atmospheric circulation of the region.
- *The cause of mountain waves is Differences in air pressure over a mountain range.*

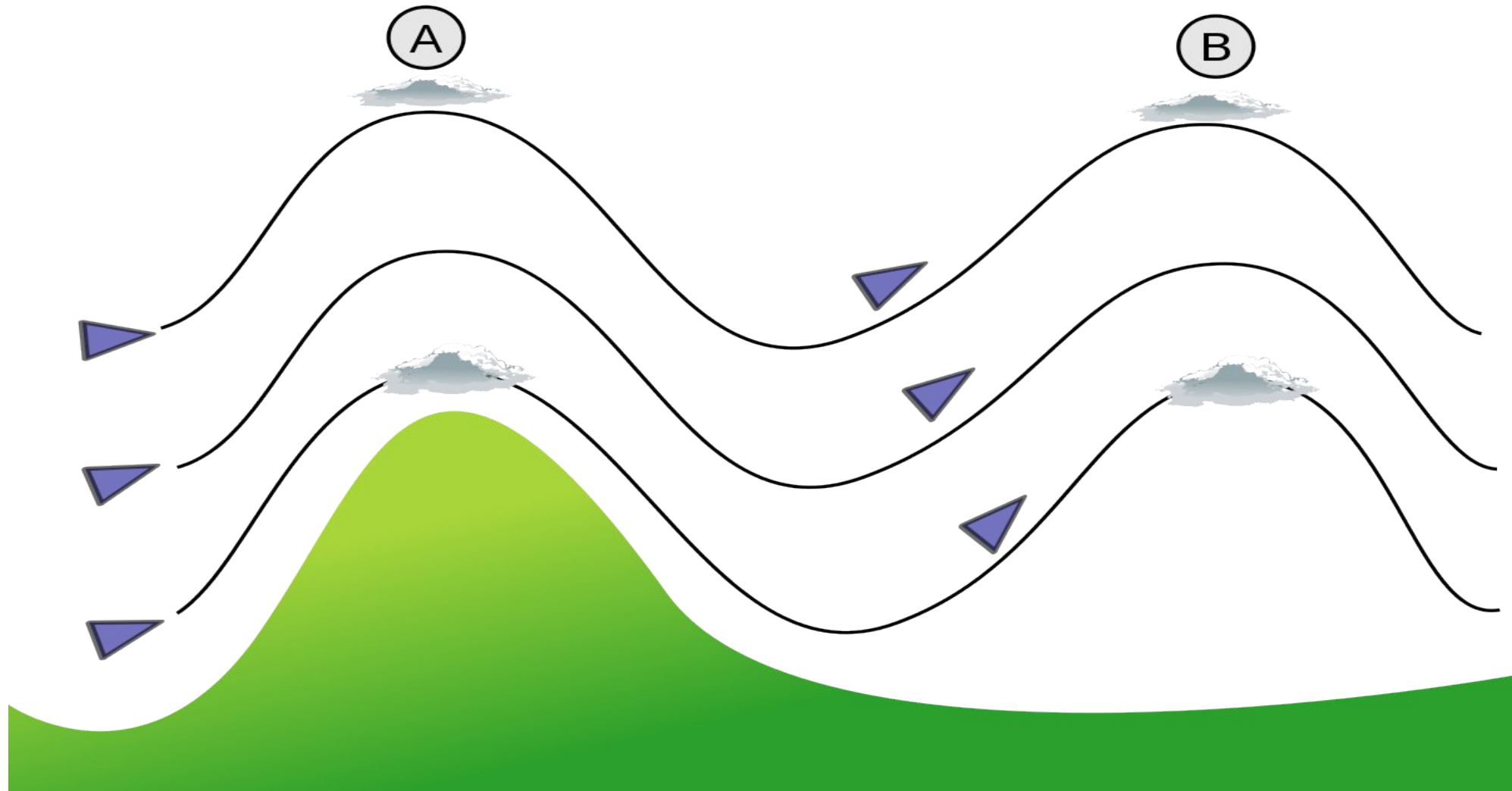
Lee Waves

- Lee waves specifically refer to a type of atmospheric wave that forms on the lee side (downwind side) of a mountain or mountain range as a result of the flow of air over the terrain.
- These waves occur downstream of the mountains, where the air descends and creates regions of sinking motion in the atmosphere.
- Lee waves are characterized by alternating regions of rising and sinking air, with the most pronounced effects typically observed in the lee of the mountains.
- They can induce clear air turbulence, which can be hazardous for aviation, and can sometimes be visually manifested by the formation of lenticular clouds or rotor clouds.
- Lee waves are a subset of mountain waves, specifically referring to the waves formed on the lee side of the mountains.
- Clouds are associated with lee waves *Cirrus clouds*.

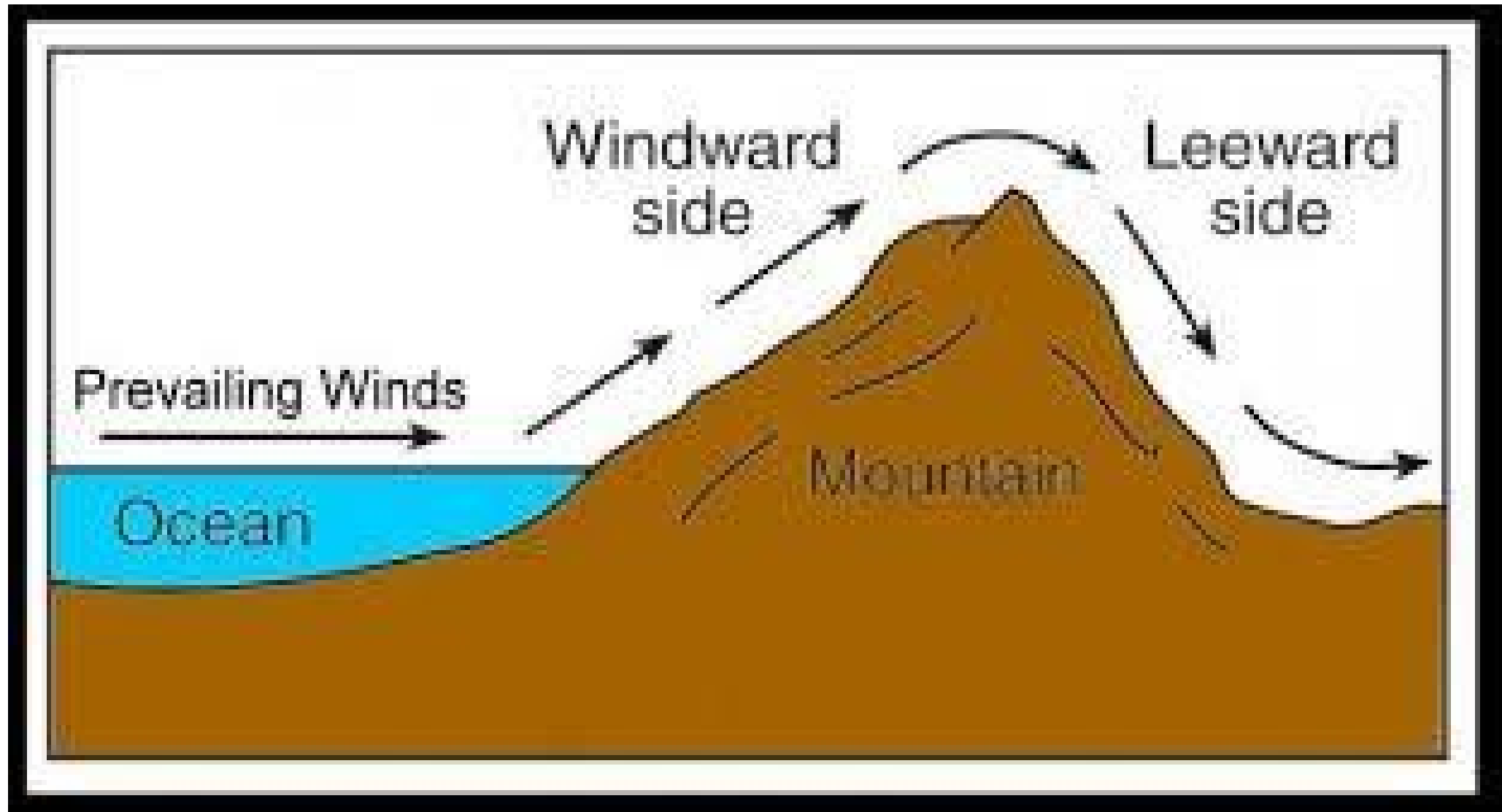
Conti.....

- Gravity waves generated by the flow over a mountain are referred to as *mountain waves* or *lee waves*.
- Mountain waves sometimes reveal their presence through dramatic cloud formations, such as smooth lenticular clouds, cap cloud and ragged rotor clouds.

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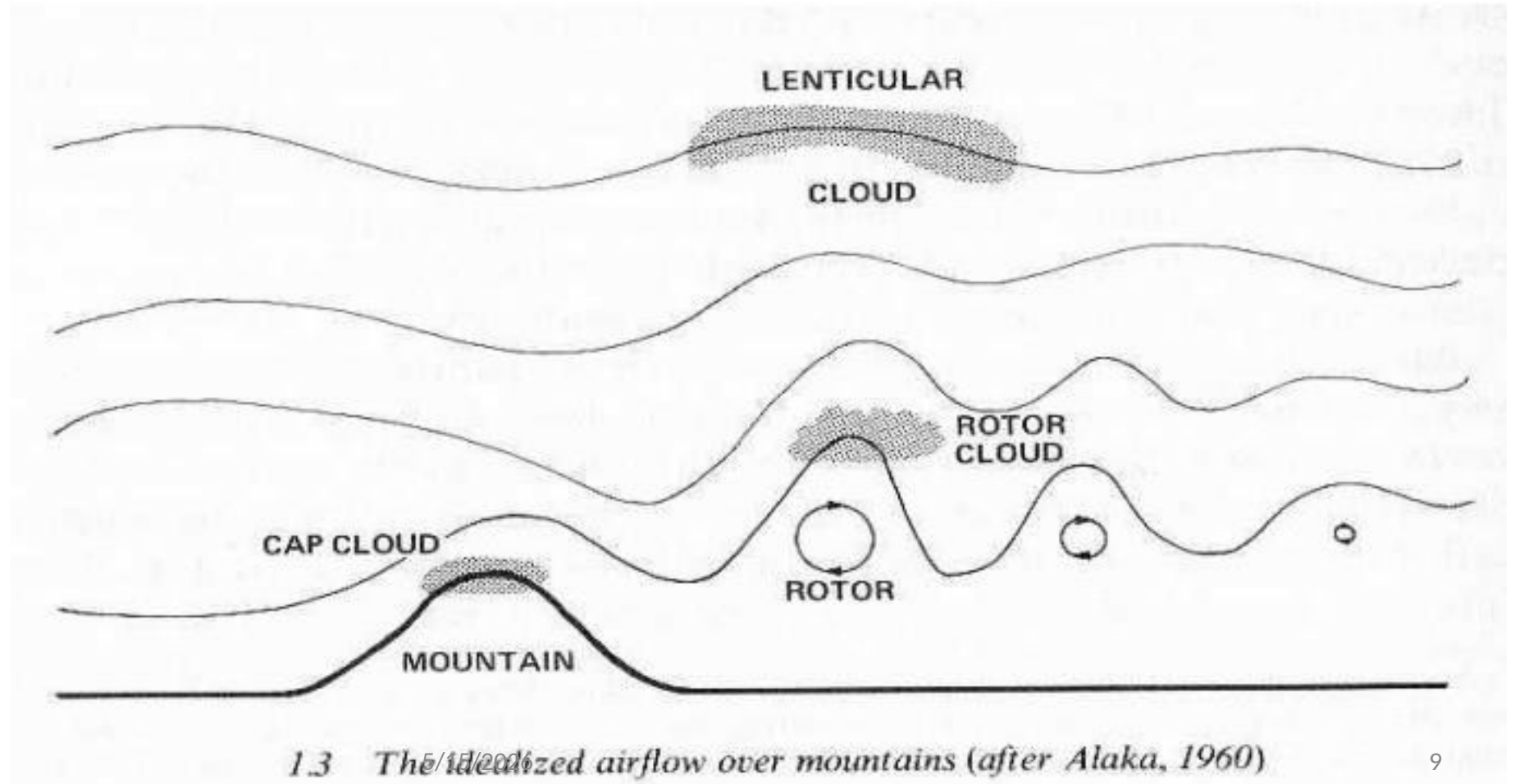


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Conti.....

- In meteorology, lee waves are atmospheric stationary waves.
- Meteorologists forecast mountain waves with a high degree of accuracy.



Formation

- Mountain waves form when stable air is forced to rise over a mountain barrier. As the air ascends, it cools adiabatically (i.e., without exchanging heat with the surrounding environment) due to decreasing pressure with height.
- If the atmospheric conditions are right, the air may become saturated, leading to the formation of clouds on the windward side of the mountain (orographic clouds).

Lee Waves:

- The descending side of the mountain, known as the lee side, is where the most pronounced mountain waves are often observed.
- Here, the descending air creates regions of sinking motion in the atmosphere, which can lead to clear air turbulence and the dissipation of clouds.

Impacts

- Mountain waves can have significant impacts on weather, particularly in terms of turbulence and aviation.
- Aircraft flying through mountain wave conditions may experience turbulence, especially in the vicinity of the crests and troughs of the waves.
- Pilots often try to avoid areas of known mountain wave activity to ensure passenger comfort and safety.

Typical Clouds

With sufficient moisture, three typical wave clouds will form:

- The **CAP CLOUD** or FOEHNWALL is mostly on the windward side of the mountain and it dissipates on the lee side where the air descends, heats by compression and can absorb large amounts of moisture.
- The second cloud type and a “sign post” of wave conditions is the **LENTICULAR clouds**. Mountain waves may or may not be marked by lenticular clouds.

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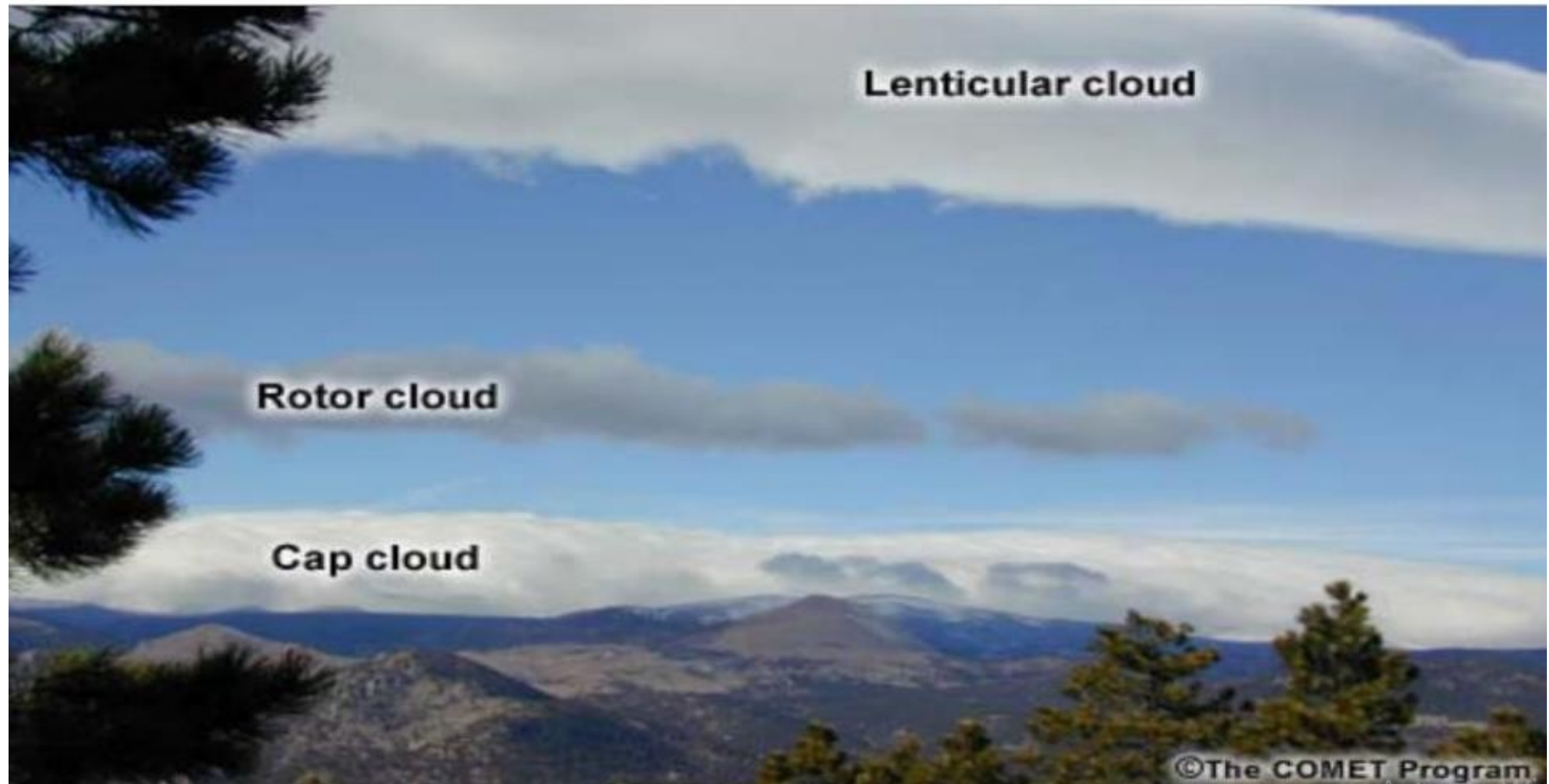
- Lenticular clouds are lens-shaped clouds that are composed of water vapor or ice crystals, normally from 20,000 to 40,000 feet. The development of lenticular clouds depends on the humidity of the air aloft

Conti.....

- As the air is forced to rise, it cools by expansion to the dew point, and if sufficient moisture is present, it condenses out and moves over the crest of the wave and down the other side until the heat of compression evaporates and reabsorbs it.
- This forms the lenticular clouds that mark wave crests.
- The third cloud type is a **ROLL, ROTOR or ARCUS** cloud.
- It may form and be nearly as stationary as the lenticular. It is formed continually on the upslope and dissipated on the downslope side. This cloud, in contrast to the lenticular, is definitely a “signpost of the sky” indicating avoidance of its area.

Conti.....

- Interaction of rising and descending air between the troughs and crests of a wave and the high velocity of the air spilling over the mountain crest like a waterfall, causing strong downdrafts, can combine with the lower velocity of air near the surface to cause the rotor.
- **The rotor cloud moves** in a rotary motion that may or may not be detected by the pilot. Many time the rotor cloud looks much like a “fair weather cumulus” or a patch of innocuous stratocumulus unless you take a closer look.



Mesoscale Vortices

Mesoscale Vortices

What are they?

- Small-scale rotating systems
- Diameter: tens to hundreds of km

Examples

- Mesoscale Convective Vortices (MCVs)
- Small tropical disturbances

Importance in the Tropics

These systems are very active in the tropics because of:

- Strong solar heating
- High moisture
- Complex terrain (e.g., East Africa, Southeast Asia)

Importance

- Can:
 - Enhance rainfall
 - Trigger storms
 - Develop into larger systems

Land and sea breezes

Direction of Wind Flow:

Sea Breeze:

- ✓ Sea breezes blow from the sea towards the land.
- ✓ During the day, the land heats up faster than the water, creating a temperature difference that results in lower pressure over the land.
- ✓ As a result, air flows from the higher pressure over the sea towards the lower pressure over the land, generating the sea breeze.

Land Breeze:

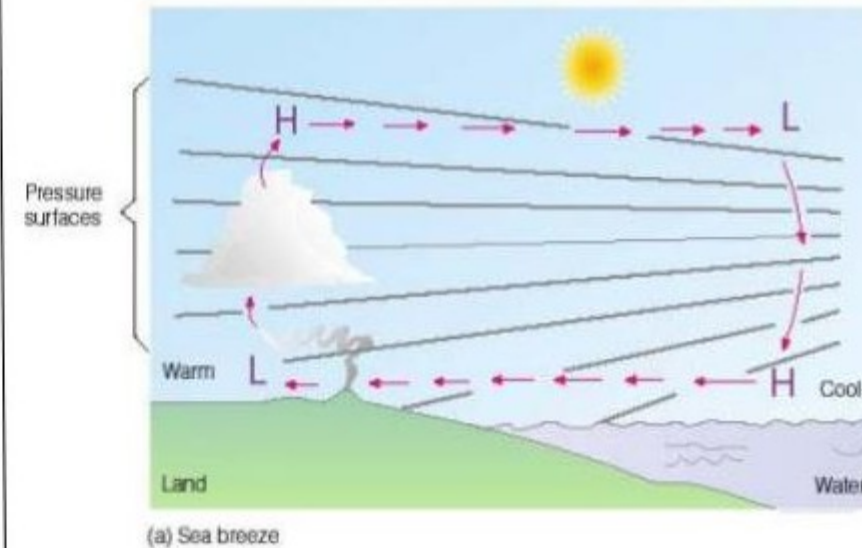
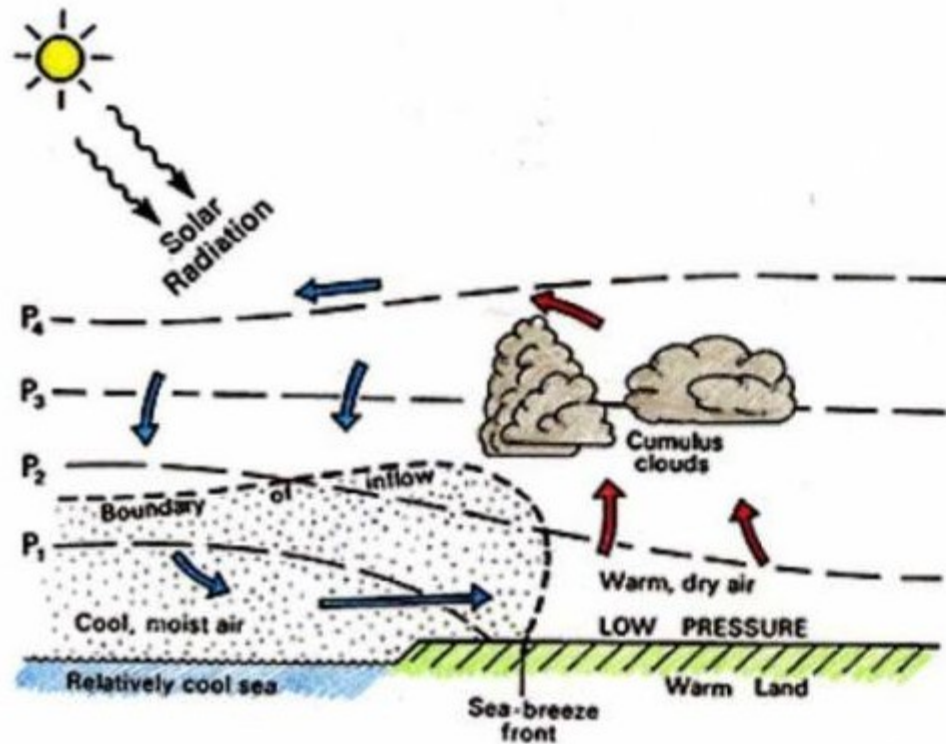
- ✓ Land breezes blow from the land towards the sea.
- ✓ At night, the land cools down more rapidly than the water, causing the air above the land to become cooler and denser compared to the warmer air over the water.
- ✓ This creates a pressure gradient with higher pressure over the land and lower pressure over the sea, leading to the flow of air from land to sea.

Effects and Implications:

- **Sea Breeze:**
 - ✓ can moderate temperatures in coastal areas
 - ✓ bring relief from hot conditions
 - ✓ influence local weather patterns, such as the development of clouds and precipitation along the coast.
 - ✓ dispersion of pollutants inland.
- **Land Breeze:**
 - ✓ can have less noticeable effects compared to sea breezes, but they can still influence
 - ✓ **local weather conditions**, particularly near coastal regions.
They may contribute to the **transport of pollutants out to sea** and can affect the behavior of marine ecosystems.

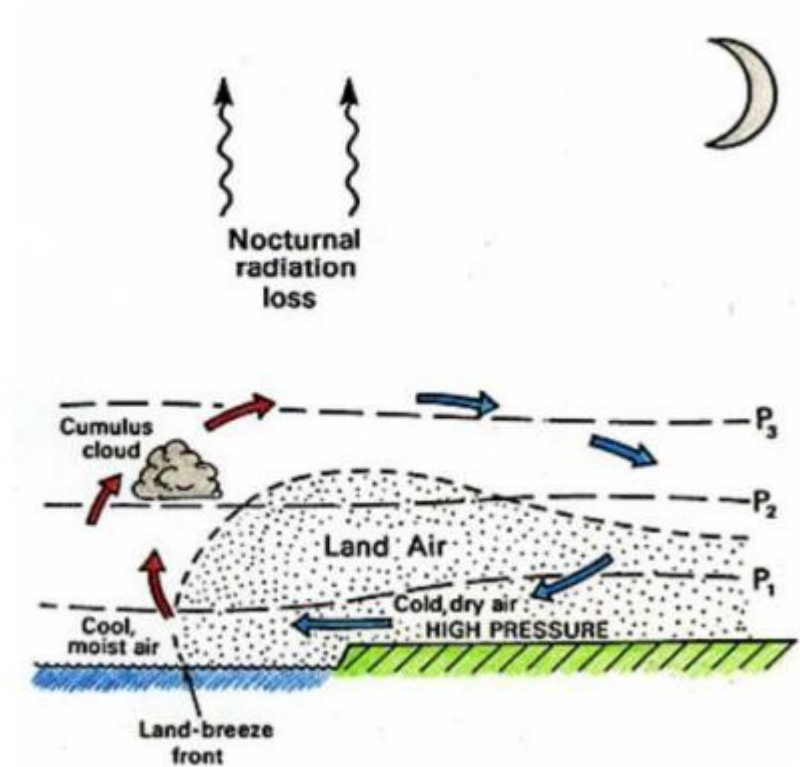
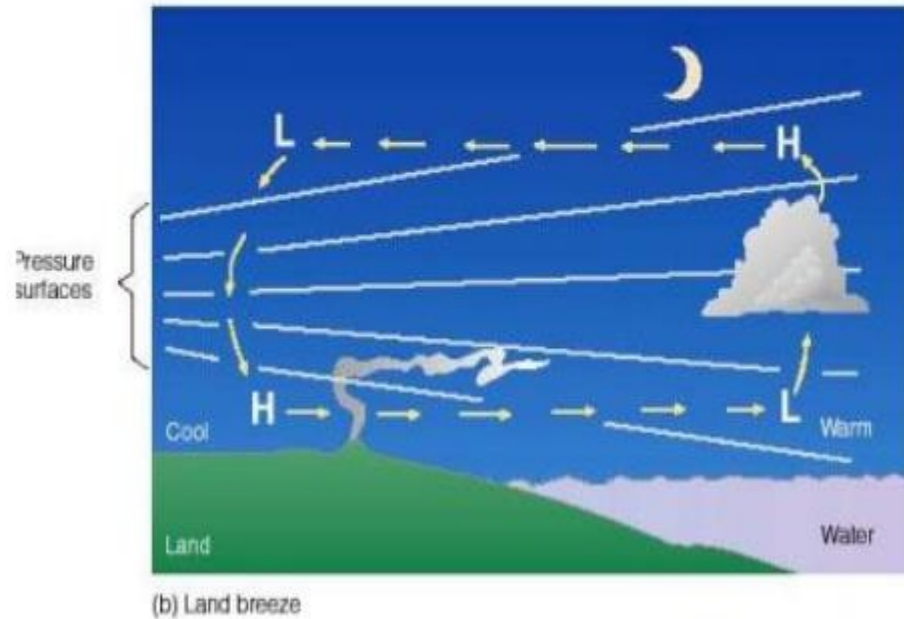
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Sea-breeze



Conti.....

Land-breeze



Conti.....

- A sea breeze develops a few hours after sunrise, continues during the daylight hours, and dies down after sunset.
- Later, a seaward-blowing land breeze appears and continues until after sunrise.
- The sea breeze may extend up to 50 or 100 km inland, but the seaward range of the land breeze is much smaller.
- In the vertical, the sea breeze reaches altitudes of 1300–1400 meters in tropical coastal areas, with a maximum speed at a few hundred meters above the ground.

Anabatic and Katabatic winds

Anabatic Winds:

- ✓ also known as upslope winds, occur when the sun heats the surface of the Earth, causing the air in contact with the surface to warm up and become less dense.
- ✓ As this warmed air rises, it flows **upslope** along the terrain, following the contours of hillsides, mountains, or other elevated features.
- Anabatic winds are typically observed during the **daytime** when solar heating is strongest, and they are more common in areas with steep slopes or mountainsides exposed to the sun.
- These winds are often gentle and can be important for local microclimates, as they may bring cooler air from higher elevations down into valleys or help to ventilate valleys and basins.
- Anabatic winds are prevalent in regions like mountainous areas, where they contribute to the circulation of air and can influence local weather patterns.

Conti.....

- When mountain slopes are heated by the sun during the day in fair-weather conditions, the warm air rises along the slope as an *anabatic wind*.
- Along a mountain crest, where anabatic winds from the two sides of the mountain meet, the air rises above the crest and can create *anabatic cumulus clouds* if sufficient moisture is present.
- Associated with these upslope winds are *return circulations* that include weak downdrafts aloft in the centers of the valley.

Katabatic Winds:

- Katabatic winds, also known as **downslope winds**, occur when **air cools** and becomes denser, typically at higher elevations or over cold surfaces such as glaciers or snowfields.
- The dense, cold air descends downslope under the influence of gravity, accelerating as it moves downhill.
- Katabatic winds are often strongest at **night when temperatures cool**, but they can also occur during the day in areas with strong cold-air drainage or when air masses with differing temperatures interact.
- These winds can be quite strong and are known for their gusty and turbulent nature, particularly as they funnel through mountain passes or valleys.
- Katabatic winds are common **in polar and high-altitude regions**, where they play a significant role in shaping local climates and can have important impacts on ecosystems, transportation, and human activities.

Conti.....

- At night when the mountain slopes are cooled by longwave radiation, the chilled air sinks downslope as a cold *katabatic wind* due to its negative buoyancy.
- This flow is compensated by a return circulation having weak rising air aloft over the valley center. The chilled air that collects as a *cold pool* in the valley floor is conducive to *fog* formation.

Conti.....

- anabatic winds flow upslope in response to daytime heating and are associated with warmer air rising along terrain features, while katabatic winds flow downslope due to the cooling and increased density of air, often occurring at night or in cold environments. Both types of winds are influenced by topography and can have notable effects on local weather and climate.

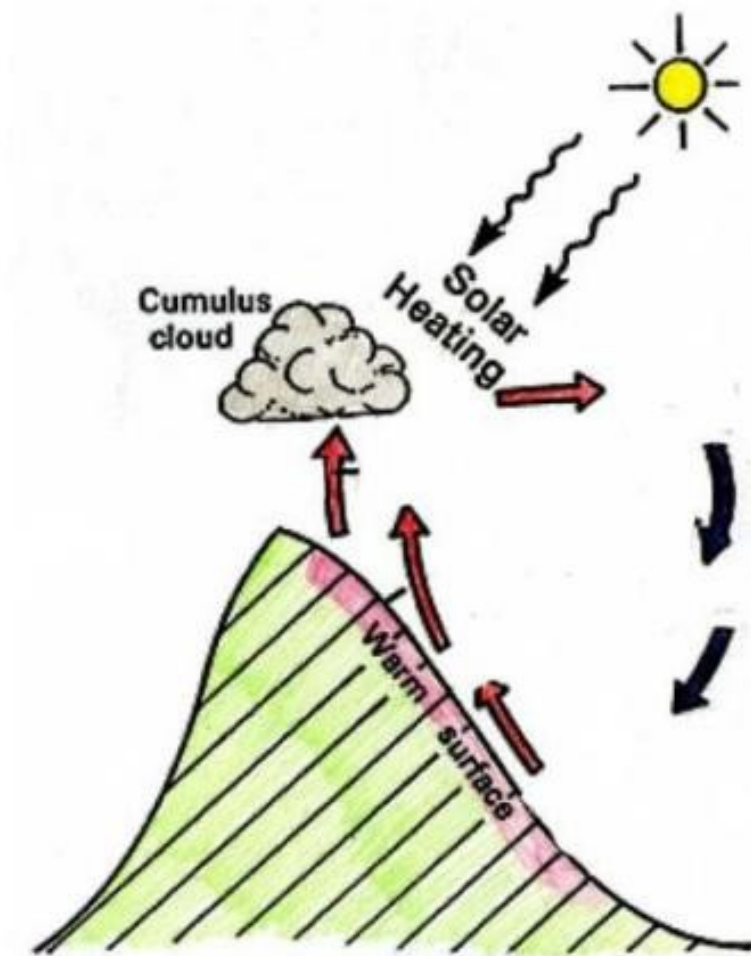
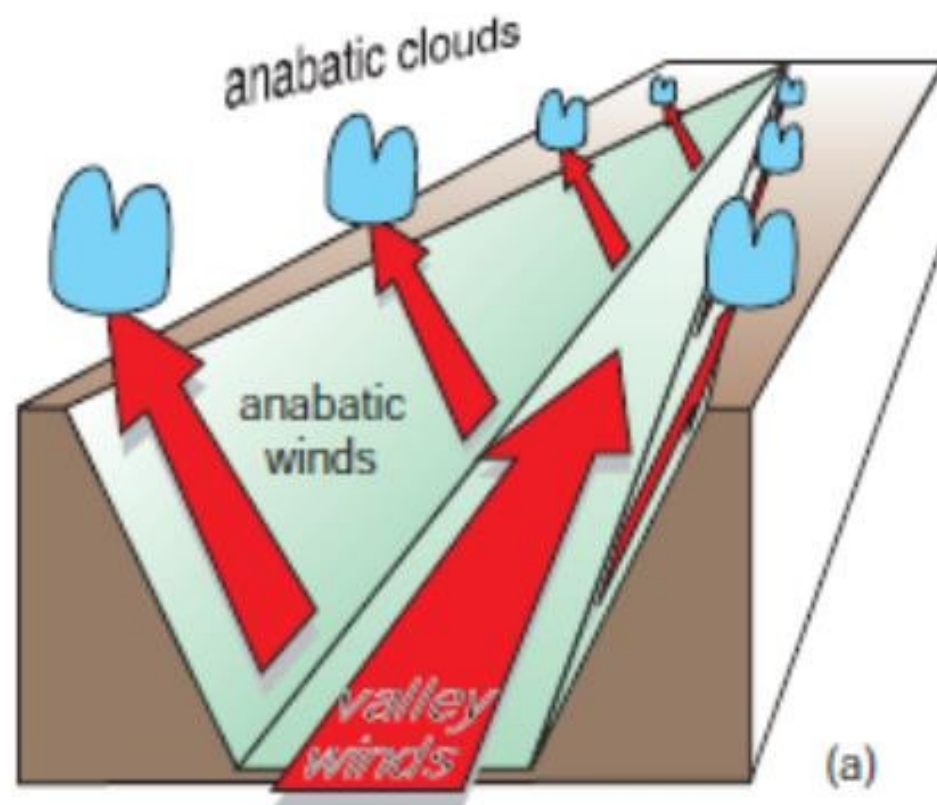
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- Anabatic and katabatic flows are examples of a broader class of diurnally varying thermally driven circulations, which also include land and sea breezes.

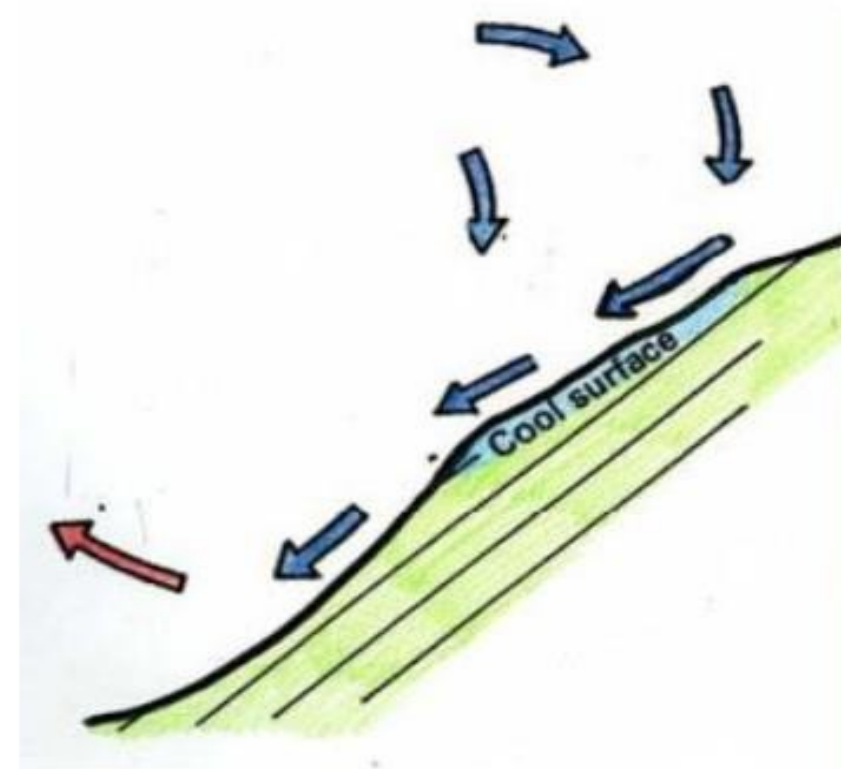
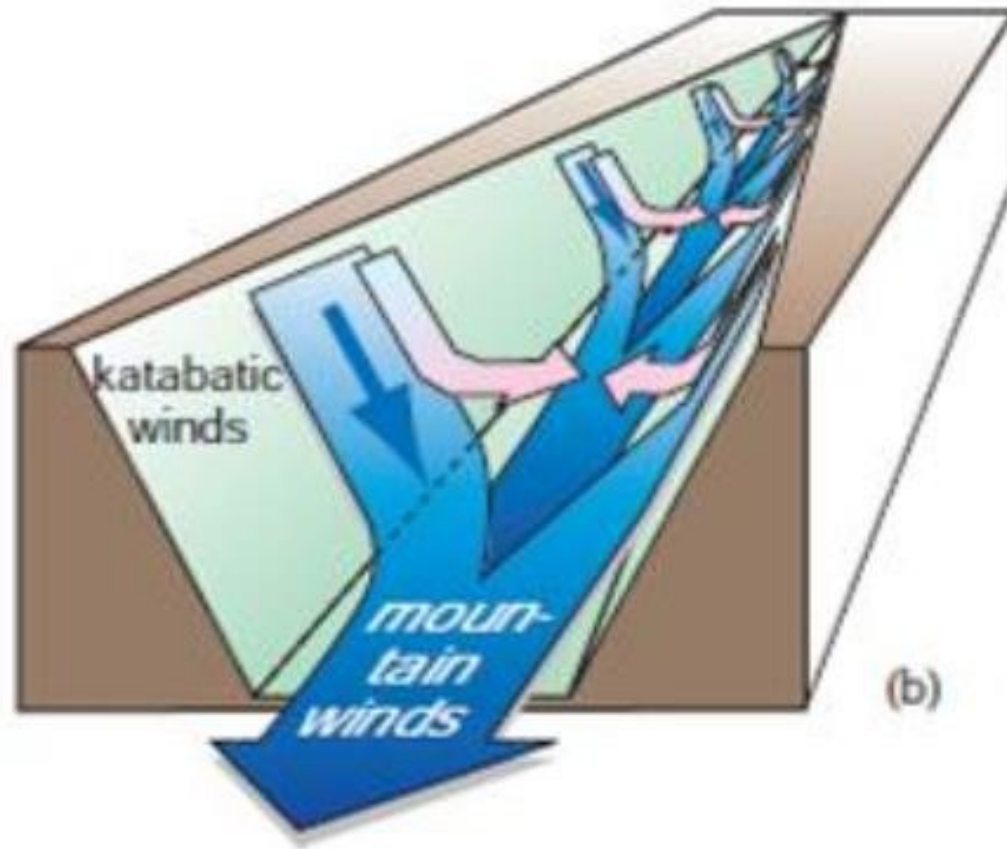
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- Horizontal temperature gradients caused by differential solar heating or longwave cooling cause horizontal pressure gradients.
- This pressure gradient force induces an upslope (anabatic) flow during daytime, while the increased density of the air just above the surface at night induces a downslope (katabatic) flow.

Anabatic winds



Katabatic Winds



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Thanks !

Tropical Meteorology



Weather Systems Affecting Ethiopia



Objective

- *To Know The Rainfall Regimes in Ethiopia*
- *To Identify the Weather Systems Affecting Ethiopia*

Ethiopian Climate

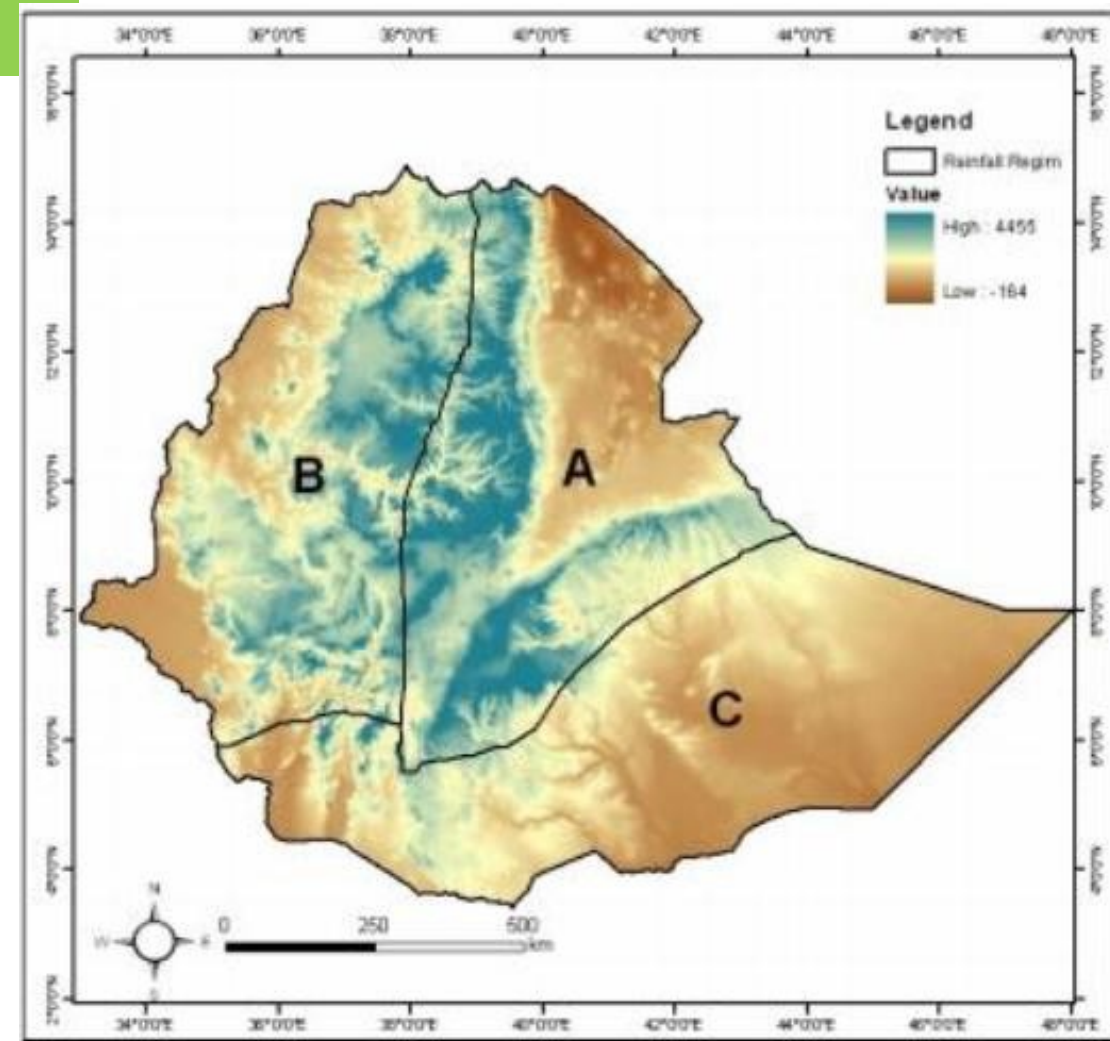
- Ethiopia is located in the Horn of Africa, with an area of 1 127 127 sq km, of which, 7444 sq km is water.
- The topography of Ethiopia is highly diverse, with elevation ranging from -125 m at the Denakil Depression to 4620 m at Ras Dejen.
- More than 45% of the country is dominated by a high plateau with a chain of mountain ranges that is divided by the East African Rift Valley.

Conti.....

- There are numerous lakes (Abaya, chamo, Awash etc) in the Great Rift Valley. This region with elevations greater than 1500 m is known as the highlands.
- Surrounding the highlands are regions known as the lowlands (<1500 m).
- Elevation differences can greatly influence local microclimate, while these differences may be partially influencing the variability and trends in regional rainfall.

Rainfall Regimes in Ethiopia

- The rainfall regime in Ethiopia is characterised as uni-modal and bi-modal systems influenced by topographical variation in the country, seasonal cycles and opposing responses to regional and global weather systems, consequently, three rainfall regimes are commonly identified.



Regime A

- That comprises the central and the eastern part of the country has a bimodal rain classified as the long rainy season (June–September) and short rains (March-May) locally referred as *Kiremt* and *Belg* rains respectively.
- The rest of the months (October to February) are dry period.

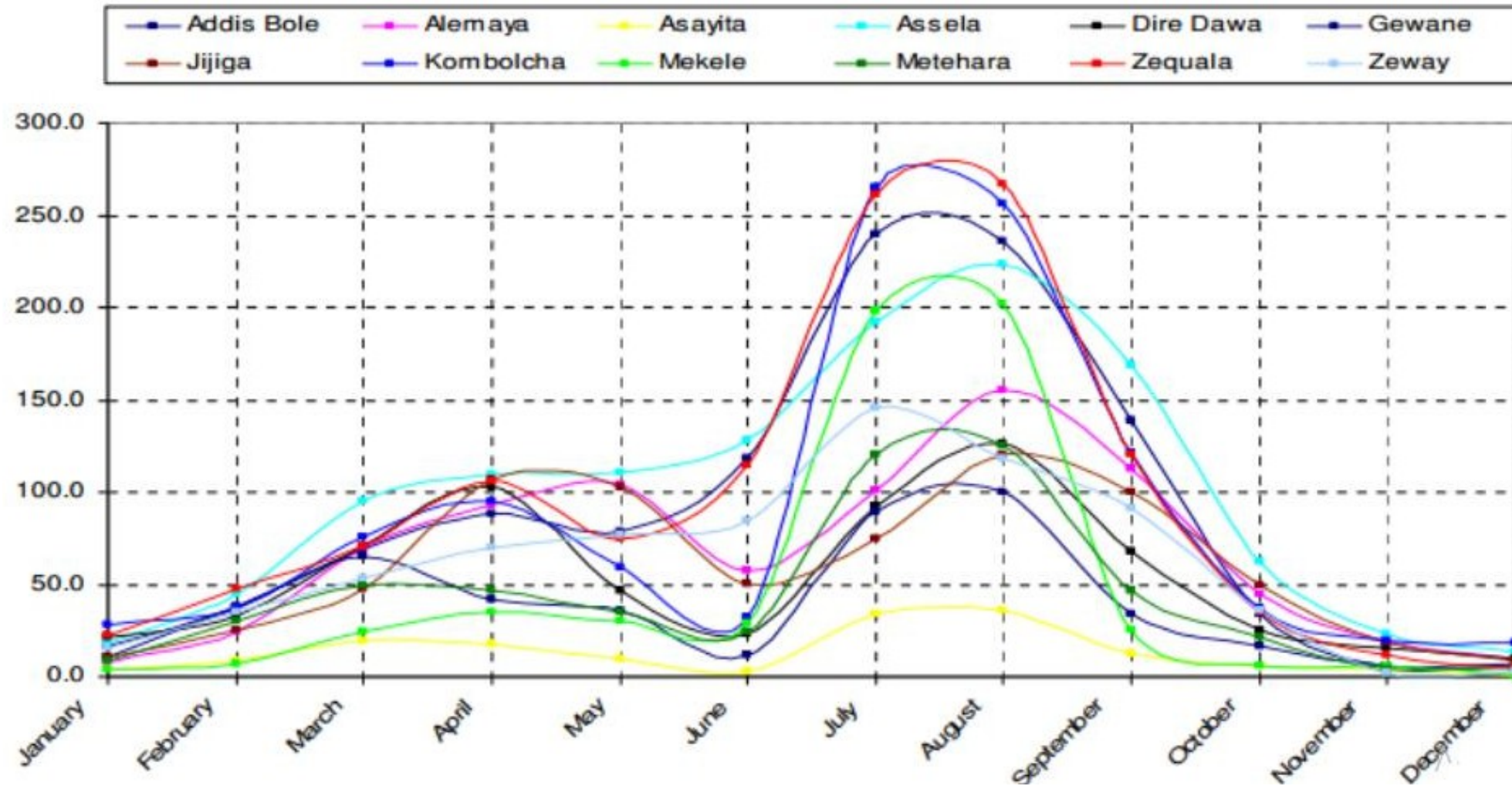
Regime B

- In the western part of the country (from southwest through to northwest) has a mono-modal rainfall pattern, and the rainy period ranges from jun through September mainly in the western and south-western part of the country, and decreases northwards.

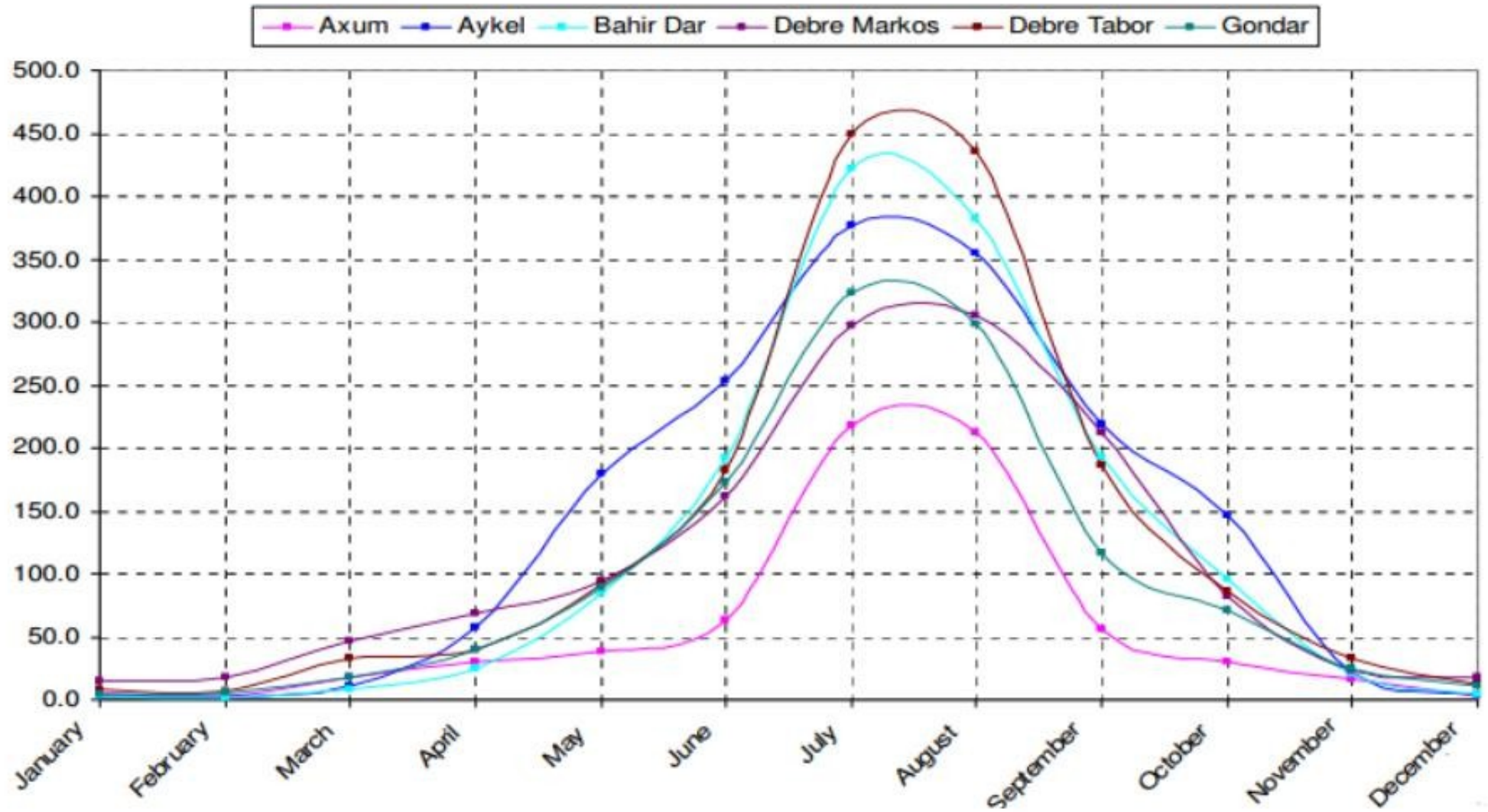
Regime C

- That comprises the south and south-eastern part of Ethiopia has two distinct wet and dry seasons.
- The main rain season is from February through May, and short rains from October to November, and the dry periods are June to September and December to February.

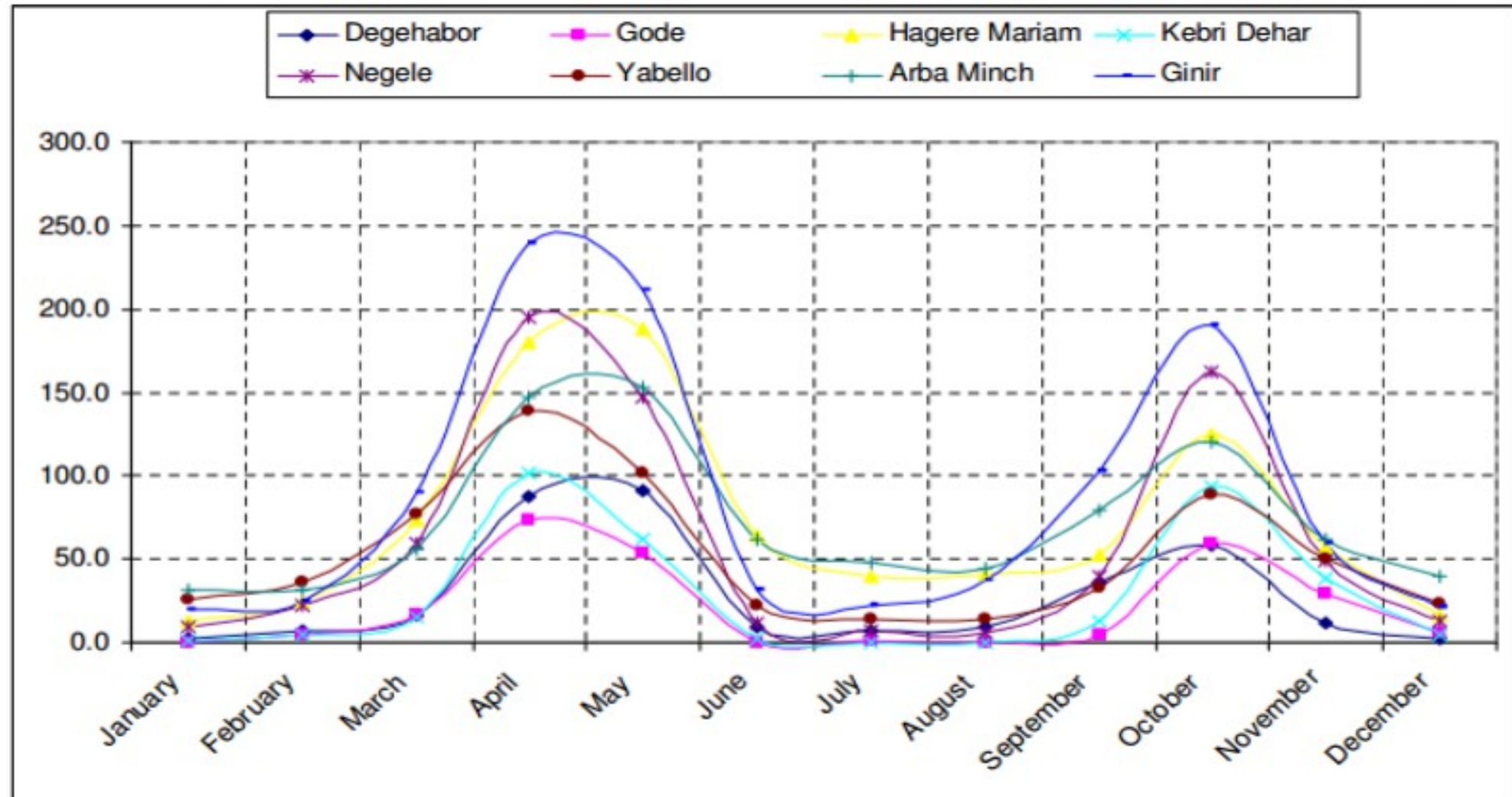
Mean monthly rainfall distribution for Central and Eastern Ethiopia (Region A)



Mean monthly rainfall distribution for the western Ethiopia (Region B1)



Mean monthly rainfall distribution for Southern and South-eastern Ethiopia (Region C)

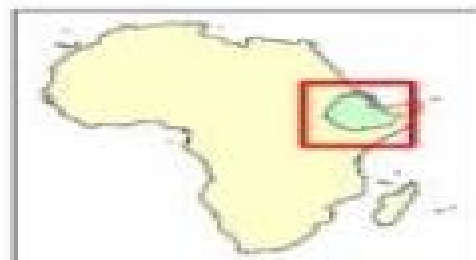


Regional Weather Features Affecting The Climate System In Ethiopia

- Ethiopia's varied topography has created three climatic zones, which have been known since ancient times as the dega, the weina dega, and the kolla.
- The dega (also known as the cool zone) occupies the central sections of the western and eastern parts of the northwestern plateau.
- The elevation of this region is mostly above 2400 m, and daily temperatures range from near freezing to 16°C.

Conti.....

- The weina dega or the temperate zone ranges between 1500 m and 2400 m in elevation, and consists of parts of the central plateau. It has daily temperatures between 16 and 30°C.
- The kolla or hot zone generally comprises areas lower than 1500 m in elevation.
- Which comprises of the Denakil Depression, and the tropical valleys of the Blue Nile.



**Legend
Climatic Zones**

- Kolla
- Weina Dega
- Dega

Conti.....

- Within each climatic zone, seasonal variations and atmospheric pressure systems contribute to the creation of three seasons, which are known as the Kiremt, Belg, and Bega.
- The Kiremt season is the main rainy season and usually lasts from June to September, covering all of Ethiopia except the southern and southeastern parts.
- The Belg season is the light rainy season and usually lasts from March to May; it is the main source of rainfall for the water deficient southern and southeastern parts of Ethiopia.
- The Bega season is the dry season and usually lasts from October to February, during which the entire country is dry, with the exception of occasional rainfall that is received in the central sections.

Weather Systems Affecting Ethiopia

The Intertropical Convergence Zone (ITCZ):

- ❑ A major driver of Ethiopia's seasonal rainfall.
- ❑ It shifts northward and southward throughout the year, influencing the rainy and dry seasons.
- ❑ Brings heavy rains during its northward movement (June–September).

Conti.....

❑ Tropical Easterly Jet (TEJ):

- Strengthens during summer, enhancing convection and rainfall over Ethiopia.

❑ Somali Jet:

- Brings moisture to southeastern Ethiopia during the Belg season.

Conti.....

- The rainy seasons in Ethiopia are influenced by different global and regional rain-bearing factors.
- The main features that affect the *Kiremt* rain include the ITCZ, Tropical Easterly Jet (TEJ), South Atlantic Ocean and South West Indian Ocean anticyclone, East African Low Level Jet (EALLJ) or Somali Jet and ENSO.
- The global and regional weather features that affect the *Belg* rain includes the ITCZ, Subtropical Westerly Jet (SWJ) stream, Arabian High, the frequency of tropical cyclones over the Southwest Indian Ocean and ENSO.

El Niño in Ethiopia

- In the case of the eastern and central equatorial Pacific Ocean, meteorologists use El Niño to predict rainfall, which is the increase in the sea surface temperatures (SST) in that part of the world.
- This temperature change results in differences in the patterns and directions of winds and ocean currents in the region, which has potentially catastrophic effects.

Conti.....

- Most studies report a strong correlation between the El Niño phenomenon and rainfall patterns in Africa.
- For the timetable of El Niño and drought and famine occurrences in Ethiopia, see Table below.

Chronology of El Niño and drought/famine in Ethiopia

El Niño Years	Drought/Famine	Regions
1539-41	1543-1562	Hararghe
1618-19	618	Northern Ethiopia
1828	1828-29	Shewa
1864	1864-66	Tigray and Gondar
1874	1876-78	Tigray and Afar
1880	1880	Tigray and Gondar
1887-89	1888-1892	Ethiopia
1899-1900	1899-1900	Ethiopia
1911-1912	1913-1914	Northern Ethiopia
1918-19	1920-22	Ethiopia
1930-32	1932-1934	Ethiopia
1953	1953	Tigray and Wollo
1957-1958	1957-1958	Tigray and Wollo
1965	1964-66	Tigray and Wollo
1972-1973	1973-1974	Tigray and Wollo
1982-1983	1983-1984	Ethiopia
1986-87	1987-1988	Ethiopia
1991-92	1990-92	Ethiopia
1993	1993-94	Tigray and Wollo

El Niño impacts on rainfall in Ethiopia

- El Niño–Southern Oscillation (ENSO) have an impact on a seasonal shifting of the normal rainy seasons in some regions, as a result a shortening or lengthening of the rainy seasons, particularly over tropical regions.
- In line with this, there could be a significant teleconnection linkage between ENSO and the Ethiopian rainy season.

Conti.....

- The correlation showing that rainfall could be below average through El Niño episode further more high drought probabilities during strong El Niño years whereas, La Niña events favored further temporal expansion of seasonal activities beyond the normal duration of the rainy season over a region.
- El Nino is associated with below-normal rainfall in most of Ethiopia (exception: the southeast lowlands), while La Nina linked with above normal rainefall in some regions .

Conti.....

- Subtropical high-pressure system is the main weather system affecting Ethiopia during the dry season.

Drought in Ethiopia

Causes:

- Failure of seasonal rains due to shifts in the ITCZ or anomalies in global climate systems like ENSO.
- Weakening of the Tropical Easterly Jet or Somali Jet.

Impacts:

- Severe droughts lead to water shortages, agricultural losses, and famine.

Notable Events:

- The 1984-85 famine caused by prolonged drought and poor rainfall.

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Thanks !